

The AIVI Framework:

A Measurement Model for Brand Visibility in Large Language Models

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Abstract

The emergence of Large Language Models (LLMs) has fundamentally changed how users discover information. Instead of retrieving ranked hyperlinks, modern AI systems synthesize responses by integrating information from multiple sources and recommending entities directly within generated text. Consequently, traditional search visibility metrics—including keyword rankings, backlink profiles, and click-through rates—no longer provide sufficient insight into whether a brand is discoverable inside AI-generated answers.

This paper introduces the Artificial Intelligence Visibility Index (AIVI), a measurement framework designed to quantify brand visibility across LLM-based retrieval environments. Rather than evaluating websites alone, AIVI measures how consistently an entity is recognized, cited, recommended, and semantically represented by generative AI systems.

The proposed framework decomposes AI visibility into six measurable dimensions: Entity Recognition Rate (ERR), Visibility Recall (VR), Citation Consistency (CC), Recommendation Confidence (RC), Prompt Robustness (PR), and Cross-Model Stability (CMS). These dimensions are combined into a normalized composite score that enables quantitative comparison across different language models and prompt variations.

Unlike conventional SEO metrics, AIVI focuses on entity-centric visibility within generative retrieval systems, providing a practical methodology for evaluating brand discoverability in conversational AI. The paper defines the conceptual architecture of the framework, presents a scoring methodology, discusses implementation considerations, and outlines validation strategies for future empirical research.

The framework is introduced as a measurement model and is accompanied by a small feasibility pilot across three language models and six entities. The pilot demonstrates that all dimensions except Citation Consistency can be computed from collected responses, and surfaces two boundary conditions that any implementation must handle. The paper aims to establish a common vocabulary for evaluating visibility in the emerging field of Generative Engine Optimization (GEO) and to support future standardization efforts in AI-native search measurement.

Keywords: Generative Engine Optimization (GEO), Artificial Intelligence Visibility Index, Large Language Models, Entity Retrieval, AI Search, Brand Visibility, Information Retrieval, Retrieval-Augmented Generation, Search Evaluation

1. Introduction

1.1 Background

For over two decades, digital visibility has been largely defined by search engine rankings. Search Engine Optimization (SEO) developed around optimizing webpages for ranking algorithms that produced ordered lists of hyperlinks. Visibility was primarily measured through keyword positions, organic traffic, impressions, backlinks, and click-through rates.

The rapid adoption of Large Language Models (LLMs) introduces a fundamentally different retrieval paradigm. Systems such as conversational AI assistants no longer present users with ranked documents alone. Instead, they generate synthesized responses that recommend products, organizations, services, experts, and concepts directly within natural language.

This transition shifts the optimization target from documents to entities. A company may possess strong search rankings while remaining absent from AI-generated recommendations. Conversely, an entity with modest traditional SEO performance may appear consistently in LLM responses because of strong semantic representation across trusted information sources.

These observations suggest that classical SEO metrics capture only part of modern digital visibility.

1.2 Problem Statement

Despite rapid advances in Generative Engine Optimization (GEO), there is currently no widely accepted quantitative framework for measuring how visible an entity is within LLM-generated responses.

Existing SEO metrics were designed for hyperlink retrieval systems and therefore do not directly evaluate:

- entity recognition by language models,
- recommendation frequency,
- citation consistency,
- robustness across prompt variations,
- consistency across different AI systems.

Without standardized measurements, organizations cannot objectively evaluate AI visibility or compare optimization strategies across models.

1.3 Research Objective

This paper proposes the Artificial Intelligence Visibility Index (AIVI), a measurement framework intended to quantify entity visibility inside generative AI systems.

Specifically, the objectives are to:

- define measurable dimensions of AI visibility;
- propose a normalized composite scoring model;
- describe a framework for evaluating visibility across multiple LLMs;
- establish terminology for future GEO research;
- provide a foundation for empirical benchmarking.

1.4 Contributions

This paper makes five primary contributions:

- 1. It formalizes AI visibility as a measurable construct distinct from conventional SEO performance.
- 2. It introduces the Artificial Intelligence Visibility Index (AIVI), a multidimensional framework for evaluating entity visibility across LLMs.
- 3. It defines six measurable dimensions that collectively characterize AI visibility.
- 4. It proposes a normalized composite scoring methodology suitable for cross-model comparisons.
- 5. It establishes a conceptual basis for future empirical research and standardization efforts in Generative Engine Optimization.

2. Related Work

2.1 Search Visibility in Traditional Information Retrieval

For more than two decades, web visibility has been evaluated through ranking-based search engines. Classical Search Engine Optimization (SEO) focuses on improving a webpage's position within search engine results pages (SERPs). Common performance indicators include keyword rankings, backlinks, organic traffic, click-through rate (CTR), domain authority, and page authority.

These metrics assume that users interact with ranked lists of hyperlinks and make navigation decisions based on search results. Consequently, visibility is fundamentally document-centric: a webpage is considered visible if it appears prominently in search results.

This evaluation paradigm has supported the development of mature optimization techniques and analytics platforms. However, its assumptions become less applicable in conversational AI environments, where users often receive synthesized answers instead of document lists.

2.2 Retrieval in Large Language Models

Modern Large Language Models (LLMs) introduce a substantially different information retrieval paradigm.

Rather than retrieving and presenting ranked webpages directly, LLM-based systems generate responses by integrating knowledge from model parameters and, in many deployments, external retrieval mechanisms such as Retrieval-Augmented Generation (RAG). The resulting answer may recommend entities, summarize information from multiple sources, or omit explicit references altogether.

In this setting, visibility is influenced not only by document relevance but also by factors such as:

- entity recognition,
- semantic consistency,
- source credibility,
- retrieval coverage,
- response generation behavior.

As a result, traditional SEO metrics provide only an indirect indication of whether an entity will actually appear in AI-generated responses.

2.3 Evaluation of Retrieval-Augmented Generation

Recent research on Retrieval-Augmented Generation has primarily focused on system performance rather than entity visibility.

Existing evaluation frameworks commonly assess:

- retrieval precision,
- retrieval recall,
- answer faithfulness,
- hallucination rate,
- citation accuracy,
- groundedness,
- factual consistency.

These metrics evaluate the quality of AI-generated responses from the perspective of system correctness.

However, they generally do not quantify whether a specific organization, product, researcher, or brand is consistently discoverable across prompts and across different language models.

Consequently, current RAG evaluation methodologies address response quality but not AI visibility as an independent measurement problem.

2.4 Emergence of Generative Engine Optimization

The growing adoption of conversational AI has led to the emergence of Generative Engine Optimization (GEO), an area concerned with improving the likelihood that entities appear within AI-generated responses.

Current GEO practices typically emphasize:

- structured content,
- entity clarity,
- knowledge graph alignment,
- semantic authority,
- machine-readable documentation,
- source trustworthiness.

Although these practices provide practical optimization guidance, the field currently lacks a standardized methodology for quantitatively evaluating optimization outcomes.

Most published discussions remain descriptive or operational rather than proposing formal measurement models.

2.5 Existing Measurement Limitations

Existing evaluation approaches exhibit several limitations when applied to AI-native search environments.

First, conventional SEO metrics measure webpage performance rather than entity representation.

Second, RAG evaluation frameworks measure system quality rather than discoverability.

Third, benchmark datasets usually evaluate language model capabilities instead of brand visibility.

Finally, existing GEO methodologies rarely define reproducible numerical metrics that enable comparison across models, prompts, or time.

These limitations create a measurement gap between optimization activities and observable AI visibility.

2.6 Research Gap

The literature indicates an increasing recognition that conversational AI changes how information is discovered and consumed. Nevertheless, no broadly adopted framework currently integrates entity recognition, recommendation frequency, citation behavior, prompt robustness, and cross-model consistency into a unified visibility metric.

This gap motivates the development of the Artificial Intelligence Visibility Index (AIVI).

Rather than replacing existing SEO or RAG evaluation methods, AIVI is proposed as a complementary framework specifically designed to measure entity visibility within generative AI systems.

The framework presented in this paper seeks to provide:

- a multidimensional representation of AI visibility,
- a normalized scoring methodology,
- cross-model comparability,
- reproducible evaluation procedures,
- a common terminology for future GEO research.

3. Conceptual Foundations of AI Visibility

3.1 From Search Visibility to AI Visibility

The evolution of information retrieval has fundamentally changed the meaning of digital visibility.

In traditional search engines, visibility is primarily determined by the ranking position of webpages within a list of search results. Users actively evaluate multiple hyperlinks before selecting a source of information. Consequently, search visibility has historically been synonymous with document discoverability.

Large Language Models (LLMs) alter this interaction paradigm. Rather than presenting ranked webpages, conversational AI systems synthesize information into a single response. The user often consumes the generated answer without visiting the underlying sources.

This transition changes the optimization target from documents to entities.

An organization, product, researcher, or service becomes visible only if the language model decides to include it in the generated response.

Therefore, AI visibility cannot be defined solely by webpage rankings. It must instead describe the probability that an entity appears within generated answers under relevant information-seeking contexts.

3.2 Defining AI Visibility

This paper defines AI Visibility as follows:

AI Visibility is the measurable likelihood that a specific entity is recognized, selected, and presented by a generative AI system in response to relevant user queries.

This definition introduces several important characteristics.

First, visibility is probabilistic rather than deterministic. Different prompts, models, retrieval systems, and temporal updates may produce different outputs.

Second, visibility is entity-centric rather than document-centric. The primary object of measurement is the entity itself rather than any individual webpage.

Third, visibility is contextual. An entity may demonstrate high visibility within one knowledge domain while remaining largely absent in another.

Accordingly, AI visibility should be interpreted as a multidimensional phenomenon rather than a binary outcome.

3.3 Entity-Centric Information Retrieval

Traditional search engines retrieve documents.

Generative AI systems retrieve, reason about, and synthesize information regarding entities.

This distinction is significant.

A company may own thousands of indexed webpages yet never appear in AI-generated recommendations if its entity representation is weak or fragmented.

Conversely, an organization with relatively modest web presence may consistently appear in generated answers because its semantic identity is well established across authoritative sources.

This shift suggests that entity representation constitutes a primary determinant of discoverability within LLM-based environments.

Accordingly, measurement frameworks should evaluate the entity rather than individual documents.

3.4 The AI Visibility Pipeline

This paper conceptualizes AI visibility as a sequential decision process.

An entity must successfully pass several stages before appearing in a generated response.

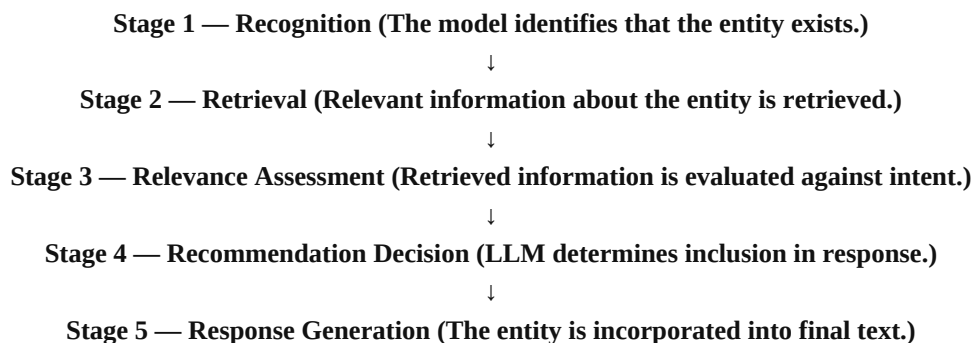


Figure 1. Sequential stages of the AI visibility decision process.

Failure at any stage may prevent the entity from appearing, even if it is highly authoritative elsewhere.

This pipeline emphasizes that visibility emerges from a chain of probabilistic decisions rather than from a single retrieval event.

3.5 Dimensions of AI Visibility

Based on the conceptual pipeline, this paper models AI visibility through six complementary dimensions.

Entity Recognition Rate (ERR)

Measures whether the language model recognizes the entity across relevant prompts. Recognition represents the prerequisite for all subsequent stages.

Visibility Recall (VR)

Measures how frequently the entity appears within generated responses when it is relevant. This dimension estimates discoverability rather than ranking performance.

Citation Consistency (CC)

Measures whether supporting sources remain stable across repeated interactions. High consistency indicates stronger informational grounding.

Recommendation Confidence (RC)

Measures the degree to which the model explicitly recommends or endorses the entity instead of merely mentioning it. Recommendation represents a stronger form of visibility than simple occurrence.

Prompt Robustness (PR)

Measures the stability of visibility across different prompt formulations expressing the same intent. High robustness indicates resilience against prompt variation.

Cross-Model Stability (CMS)

Measures consistency across multiple LLM providers. An entity that appears consistently across different models demonstrates higher AI visibility than one appearing in only a single ecosystem.

3.6 A Multidimensional Measurement Perspective

No single metric sufficiently captures AI visibility. For example:

- An entity may be recognized by every model but never recommended.
- It may receive recommendations only for narrowly phrased prompts.
- It may appear frequently in one model but be absent from others.
- It may be cited inconsistently across repeated evaluations.

These scenarios illustrate that AI visibility cannot be represented adequately through a one-dimensional indicator.

Instead, the construct should be viewed as a multidimensional latent variable inferred from multiple observable measurements.

The AIVI framework therefore combines several complementary indicators into a unified evaluation model.

3.7 Conceptual Model

The proposed framework can be summarized as follows:

**Entity → Recognition (ERR) → Retrieval → Visibility (VR) → Citation (CC) → Recommendation (RC) →
Prompt Robustness (PR) → Cross-Model Stability (CMS) → AIVI**

Figure 2. The AI visibility pipeline.

4. AIVI Framework Architecture

4.1 Design Principles

The Artificial Intelligence Visibility Index (AIVI) is designed as a composite measurement framework rather than a single observational metric.

Instead of relying on one indicator, AIVI combines multiple observable variables that collectively estimate the latent construct of AI visibility.

The framework follows five design principles:

- **1. Entity-Centricity:** The unit of analysis is the entity (e.g., brand, organization, product, or person), not the webpage or document.
- **2. Model Independence:** The framework is intended to be applicable across different LLM providers without depending on proprietary internal model characteristics.
- **3. Prompt Robustness:** Measurements should remain stable across semantically equivalent prompt formulations.
- **4. Reproducibility:** Independent evaluators following the same protocol should obtain comparable results.
- **5. Extensibility:** Additional dimensions may be incorporated in future revisions without altering the conceptual architecture.

These principles aim to ensure that AIVI functions as a general evaluation framework rather than a platform-specific benchmark.

4.2 Measurement Model

AI Visibility is conceptualized as a latent variable V_{AI} that cannot be observed directly. Instead, it is inferred from six observable dimensions:

$$V_{AI} = f(ERR, VR, CC, RC, PR, CMS)$$

where:

- ERR: Entity Recognition Rate
- VR: Visibility Recall
- CC: Citation Consistency
- RC: Recommendation Confidence

- PR: Prompt Robustness
- CMS: Cross-Model Stability

This formulation reflects the assumption that AI visibility emerges from multiple interacting characteristics rather than a single observable behavior.

Section 4.5 operationalizes f as a weighted linear combination of the six normalized dimension scores; the composite index AIVI is therefore the estimator of the latent variable V_{AI} adopted in this paper.

4.3 Dimension Definitions

Entity Recognition Rate (ERR)

Measures the proportion of relevant prompts in which the entity is correctly recognized.

$$ERR = \text{Recognized Prompts} / \text{Relevant Prompts} \quad (0 \leq ERR \leq 1)$$

Visibility Recall (VR)

Measures how frequently the entity appears when it is contextually relevant.

$$VR = \text{Visible Responses} / \text{Relevant Queries}$$

Citation Consistency (CC)

Measures the stability of supporting citations across repeated evaluations. Rather than counting citations alone, CC evaluates whether the underlying supporting sources remain consistent under repeated sampling.

Let S_r denote the set of sources cited in repetition r , and let R be the total number of repetitions for a given prompt. CC is defined as the mean pairwise Jaccard similarity between the cited source sets:

$$CC = [2 / (R(R-1))] \cdot \sum_{\{r < s\}} (|S_r \cap S_s| / |S_r \cup S_s|) \quad (0 \leq CC \leq 1)$$

CC is undefined when no repetition returns any citation; such cases should be reported as not applicable rather than scored as zero.

Recommendation Confidence (RC)

Measures the strength with which the model recommends the entity. Unlike binary appearance, recommendation confidence considers the qualitative role of the entity within the generated response. This dimension may be operationalized using a predefined annotation protocol or future automated classifiers.

RC is operationalized through an ordinal endorsement scale applied by annotators to each relevant response j : $c_j = 0$ if the entity is absent; $c_j = 1/3$ if it is mentioned without endorsement; $c_j = 2/3$ if it appears among several recommended options; $c_j = 1$ if it is presented as the primary or preferred choice.

$$RC = (1/N) \cdot \sum_{j=1}^N c_j \quad (0 \leq RC \leq 1)$$

Because RC depends on human judgment, studies applying this dimension should publish their annotation guidelines and report inter-rater agreement using a chance-corrected statistic such as Cohen's κ or Krippendorff's α .

Prompt Robustness (PR)

Measures visibility stability under semantically equivalent prompt reformulations. Let n represent the number of prompt variants. PR may be expressed as:

$$PR = 1 - 2\sigma(V_p) \quad (0 \leq PR \leq 1)$$

where $\sigma(V_p)$ is the population standard deviation of the per-prompt visibility scores across the n prompt variants. Because each score is normalized to $[0, 1]$, $\sigma(V_p)$ cannot exceed 0.5; the factor of 2 therefore rescales PR onto the unit interval. Higher PR indicates lower prompt sensitivity.

Cross-Model Stability (CMS)

Measures consistency across multiple LLM providers. Let V_k denote the visibility score obtained for model k across m evaluated models. CMS is defined symmetrically to PR:

$$CMS = 1 - 2\sigma(V_k) \quad (0 \leq CMS \leq 1)$$

where $\sigma(V_k)$ is the population standard deviation of the per-model visibility scores. Lower variance across models corresponds to higher stability.

4.4 Normalization

Each observable metric is normalized to the interval $[0, 1]$ before aggregation.

Normalization enables comparison among variables measured on different scales and facilitates future integration of additional dimensions.

The framework intentionally avoids hard-coded thresholds because optimal calibration may vary across datasets, languages, and application domains.

Two dimensions require a conditional reading. Because PR and CMS reward low variance, an entity that is uniformly absent across all prompts or all models yields zero variance and therefore maximal PR and CMS. These two dimensions should consequently be interpreted only jointly with ERR and VR, and reported as undefined when $VR = 0$.

4.5 Composite Index

Instead of prescribing fixed weights, the framework defines AIVI as a weighted composite score:

$$AIVI = \sum_{i=1}^6 w_i \cdot x_i \quad \text{subject to} \quad \sum_{i=1}^6 w_i = 1$$

where x_i represents a normalized dimension score and w_i denotes its corresponding weight.

This flexible formulation allows researchers to estimate weights empirically using techniques such as: Expert elicitation, Analytic Hierarchy Process (AHP), Principal Component Analysis (PCA), Confirmatory Factor Analysis (CFA), or Structural Equation Modeling (SEM) depending on available data.

In the absence of empirically estimated weights, this paper recommends equal weighting ($w_i = 1/6$) as a reporting baseline, on the grounds that the six dimensions are treated as conceptually distinct and no prior evidence currently justifies unequal weighting. Any departure from this baseline should be stated explicitly and justified.

4.6 Interpretation

The resulting AIVI score provides a relative measure of AI visibility rather than an absolute measure of authority.

Higher scores indicate that an entity is more consistently recognized, retrieved, cited, recommended, and represented across conversational AI systems.

However, the index should not be interpreted as a direct measure of quality, expertise, or factual correctness. Instead, AIVI estimates the probability of effective discoverability within AI-mediated information environments.

5. AIVI Benchmark Protocol (ABP)

5.1 Motivation

A measurement framework is only useful if independent researchers can reproduce its results.

One of the major challenges in evaluating AI visibility is the absence of standardized testing procedures. Different researchers often use different prompts, different models, different evaluation criteria, and different sampling strategies. Consequently, reported visibility scores are difficult to compare across studies.

To address this limitation, this paper proposes the AIVI Benchmark Protocol (ABP)—a standardized evaluation procedure for measuring entity visibility across Large Language Models.

Rather than prescribing a specific software implementation, ABP defines a reproducible methodological workflow.

5.2 Evaluation Pipeline

The benchmark consists of seven sequential stages:

Target Entity → Prompt Set Construction → Multi-Model Execution → Response Collection → Metric Extraction → Dimension Scoring → AIVI Calculation

Figure 3. The AIVI Benchmark Protocol evaluation pipeline.

Each stage is designed to minimize evaluator-specific variability while preserving flexibility for different research contexts.

5.3 Benchmark Inputs

The protocol requires four primary inputs.

1. Target Entity

The organization, brand, researcher, product, institution, or service whose AI visibility is being evaluated. Examples: OpenAI, Microsoft, NVIDIA, Harvard University, Tesla. The protocol itself is entity-agnostic.

2. Prompt Set

Evaluation should be conducted using multiple prompts rather than a single query. Prompt diversity reduces prompt-specific bias and provides a more representative estimate of visibility. The benchmark distinguishes five categories:

- Direct entity queries
- Comparative queries
- Recommendation queries
- Informational queries
- Problem-solving queries

Example: 'Which companies provide enterprise AI consulting?' instead of 'Tell me about Company X.'

3. Language Models

The benchmark encourages evaluation across multiple LLM ecosystems. Illustrative categories include: OpenAI models, Google models, Anthropic models, Perplexity, Microsoft Copilot, and other publicly accessible conversational AI systems. The protocol is intentionally designed to remain independent of any single provider.

4. Evaluation Iterations

Generative systems exhibit stochastic behavior. Therefore, repeated executions improve the robustness of measurements. The protocol recommends repeated sampling for each prompt under controlled conditions, with researchers documenting model version, temperature (where configurable), and evaluation date. Rather than prescribing an exact number of repetitions, the emphasis is on consistency and transparency.

5.4 Metric Extraction

For each generated response, evaluators record structured observations. An illustrative coding table is shown below.

Table 1. Structured Observations Coding Table.

Variable	Observation
Entity recognized	Yes / No
Entity mentioned	Yes / No
Recommended	Yes / No
Supporting citation present	Yes / No
Citation consistency	Stable / Variable
Response confidence	Low / Medium / High
Prompt identifier	P001–Pn
Model identifier	M001–Mm

This structured approach enables reproducible extraction of the six AIVI dimensions.

5.5 Aggregation

After all observations are collected: ERR is calculated, VR is calculated, CC is calculated, RC is estimated, PR is derived from prompt variation, and CMS is computed across models. The normalized dimension scores are then aggregated into the composite AIVI value.

5.6 Reporting Standard

To improve reproducibility, benchmark reports should include: evaluation date, model versions, prompt set description, sampling strategy, normalization procedure, weighting method, limitations, and raw observations (where permissible). Transparent reporting allows independent researchers to reproduce or extend benchmark results.

5.7 Limitations

The protocol recognizes several sources of variability: evolving language model versions, changes in retrieval systems, regional deployment differences, temporal knowledge updates, and prompt interpretation variability. Accordingly, AIVI scores should be interpreted as time-dependent measurements rather than permanent characteristics. Repeated evaluations over time are expected to provide a more reliable representation of AI visibility trends.

6. Validation Methodology

6.1 Purpose of Validation

The Artificial Intelligence Visibility Index (AIVI) is introduced in this paper as a conceptual measurement framework. Consequently, the objective of the validation methodology is not to demonstrate that AIVI has already been empirically validated, but to define a rigorous pathway through which its measurement properties can be evaluated in future studies.

Validation should determine whether the proposed dimensions collectively represent the latent construct of AI visibility and whether the framework produces reliable, reproducible, and interpretable results across different evaluation settings.

Accordingly, future empirical investigations should examine four principal properties: construct validity, reliability, reproducibility, and external validity.

6.2 Construct Validity

Construct validity addresses whether the six observable dimensions proposed in the AIVI framework adequately represent the theoretical concept of AI visibility.

The framework assumes that AI visibility is a multidimensional latent construct reflected by: Entity Recognition Rate (ERR), Visibility Recall (VR), Citation Consistency (CC), Recommendation Confidence (RC), Prompt Robustness (PR), and Cross-Model Stability (CMS).

Future studies may evaluate this assumption using exploratory and confirmatory factor analysis on sufficiently large datasets. Evidence supporting a coherent factor structure would strengthen the theoretical foundation of the framework.

6.3 Reliability

Reliable measurements should produce similar results when evaluation conditions remain comparable. Three complementary forms of reliability are particularly relevant.

Inter-Rater Reliability: Human evaluators may differ when determining whether an entity has been recommended or merely mentioned. Future implementations should therefore establish standardized annotation guidelines and assess agreement using appropriate inter-rater statistics.

Test–Retest Reliability: Language models evolve over time. Accordingly, repeated evaluations conducted under comparable conditions should exhibit acceptable stability while acknowledging expected temporal variation due to model updates.

Internal Consistency: If future empirical datasets indicate that several dimensions measure related aspects of AI visibility, internal consistency may be evaluated using established psychometric methods. However, because AIVI combines conceptually distinct dimensions, extremely high internal consistency should not necessarily be expected or interpreted as a requirement.

6.4 External Validity

The usefulness of AIVI depends on its applicability across different contexts. Future validation studies should therefore include: multiple industries, multiple languages, multiple countries, different entity types, and different LLM providers. Such diversity would improve confidence that the framework generalizes beyond a single benchmark environment.

6.5 Sensitivity Analysis

A practical measurement framework should detect meaningful differences without being excessively sensitive to minor experimental variation. Sensitivity analysis may therefore investigate: prompt reformulations, model version changes, retrieval updates, temporal drift, and multilingual evaluation.

6.6 Benchmark Dataset

Future validation would benefit from a publicly available benchmark dataset containing: predefined entities, standardized prompt collections, reference annotations, and repeated observations across multiple language models. An openly documented benchmark would enable independent replication and facilitate comparative evaluation of future AI visibility metrics.

6.7 Threats to Validity

Several factors may influence AIVI measurements independently of the actual visibility of the evaluated entity. Potential threats include: proprietary retrieval algorithms, undisclosed model updates, regional deployment differences, personalization, stochastic generation, external knowledge refresh cycles, and prompt ambiguity.

6.8 Validation Roadmap

The proposed validation process may be summarized as follows:

**Conceptual Framework → Pilot Annotation Study → Benchmark Dataset → Reliability Analysis →
Construct Validation → Cross-Model Evaluation → Longitudinal Monitoring**

Figure 4. Proposed validation roadmap.

7. Pilot Application

7.1 Purpose and Scope

This section reports a small pilot application of the framework. Its purpose is to demonstrate that the six dimensions can be computed end to end from collected model responses, and to expose the practical difficulties that arise when doing so. It is a feasibility demonstration rather than a benchmark study.

The entity sample, prompt set, and model sample are all too small to support any claim about the relative AI visibility of the products named below. The numerical values illustrate the measurement procedure; they are not evidence about the market position of any vendor.

7.2 Pilot Design

The pilot followed the AIVI Benchmark Protocol described in Section 5. Its parameters were as follows.

- Domain. Application observability and monitoring platforms, chosen because the category contains vendors spanning a wide range of expected public visibility.
- Entities (n = 6). Datadog, Grafana, New Relic, Honeycomb, SigNoz, and Coroot.
- Prompt set (n = 6). Two information-seeking intents, each expressed as three semantically equivalent paraphrases. Intent A concerned platform selection for a mid-sized engineering team; Intent B concerned reducing observability spend. No prompt named any target entity, so every appearance was an unprompted selection by the model.
- Models (n = 3). Claude Opus 5, Claude Haiku 4.5, and Claude Fable 5, each queried in a fresh context with no retrieval, no web access, and no tool use. A fourth model was planned but excluded after repeated API failures during the collection window; the exclusion is reported here rather than silently absorbed.
- Sampling. One response per prompt-model pair, giving 18 responses for the main matrix. Each pair was collected in an independent context so that repetitions and paraphrases could not contaminate one another. One model (Opus 5) was sampled a second time across the full prompt set to estimate test-retest agreement.
- Recognition probe. A separate direct query per model asked whether each of the six entities was a known product, with an explicit instruction not to guess. ERR was derived from these responses.
- Evaluation date. 25 July 2026.

The six prompts were as follows. Intent A: (A1) "Which observability platforms should a mid-sized engineering team consider adopting?"; (A2) "We're a 50-person engineering organization picking an observability stack. What are the main options?"; (A3) "Recommend observability and monitoring tools suitable for a growing engineering team." Intent B: (B1) "Our observability vendor bill has tripled this year. What alternatives should we look at?"; (B2) "Monitoring costs are out of control at our company. Which tools

are more cost-effective?"; (B3) "We need to cut our observability spend without losing visibility. What should we consider?"

7.3 Coding Protocol and Inter-Rater Agreement

Each response was coded for each entity using the ordinal endorsement scale defined in Section 4.3. Two annotators independently coded a 24-cell subsample drawn from four responses spanning all three models. Exact agreement was 22 of 24 cells ($P_o = 0.917$), Cohen's $\kappa = 0.87$, and linear-weighted $\kappa = 0.92$. Both disagreements fell between adjacent categories rather than between presence and absence, which suggests that the boundary between "mentioned" and "recommended" is the harder judgement, as anticipated in Section 6.3.

Table 2. Observation summary by entity and model (18 responses, 6 prompts $\times$ 3 models).

Table 2 reports the resulting observation summary.

Entity	ERR	Mentions (of 18)	VR Opus 5	VR Haiku 4.5	VR Fable 5
Datadog	1.00	15	0.67	1.00	0.83
Grafana	1.00	15	0.83	0.83	0.83
New Relic	1.00	13	0.33	1.00	0.83
Honeycomb	1.00	12	0.83	0.50	0.67
SigNoz	1.00	7	0.50	0.00	0.67
Coroot	0.67	0	0.00	0.00	0.00

7.4 Results

Table 3 reports the six dimension scores and the composite index. AIVI was computed using the equal-weight baseline recommended in Section 4.5, applied over the dimensions that were defined for each entity. Citation Consistency could not be computed for any entity and is reported as not applicable throughout.

Table 3. AIVI dimension scores and composite index (equal weights; CC excluded from the mean where undefined).

Entity	ERR	VR	CC	RC	PR	CMS	AIVI
Datadog	1.00	0.83	n/a	0.44	0.69	0.73	0.74
Grafana	1.00	0.83	n/a	0.56	0.53	1.00	0.78
New Relic	1.00	0.72	n/a	0.41	0.69	0.43	0.65
Honeycomb	1.00	0.67	n/a	0.44	0.37	0.73	0.64
SigNoz	1.00	0.39	n/a	0.24	0.37	0.43	0.49

Coroot	0.67	0.00	n/a	0.00	n/a	n/a	0.22
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7.5 Observations

Five observations from the pilot bear on the framework itself rather than on the entities measured. Comparisons across entities remain meaningful only when the same prompt set, model versions, normalization procedure, weighting strategy, and evaluation period are used.

- Recognition and visibility came apart. Coroot was identified as a real product by two of the three models, yet appeared in none of the 18 open-ended responses (ERR = 0.67, VR = 0.00). This is the clearest demonstration in the pilot that entity recognition is necessary but not sufficient for AI visibility, and that a single-dimension metric would have missed the distinction entirely.
- The degenerate case anticipated in Section 4.4 occurred. Because Coroot was absent from every prompt and every model, its visibility variance was zero, which would have produced maximal Prompt Robustness and Cross-Model Stability. The conditional rule was applied and both dimensions are reported as undefined. Without that rule, the least visible entity in the sample would have scored highest on two of six dimensions.
- Citation Consistency was undefined throughout. No response carried source citations, because the models were queried in a purely parametric configuration without retrieval. CC is therefore measurable only in retrieval-augmented deployments, and a complete AIVI profile requires evaluating an entity in both configurations.
- Prompt phrasing shifted visibility more than model identity for several entities. The cost-reduction intent produced substantially fewer named vendors than the platform-selection intent, and one paraphrase within that intent elicited no vendor names at all from two of the three models. Visibility measured on a single prompt would have been misleading for every entity in the sample.
- Cross-model divergence was entity-specific rather than uniform. Grafana was mentioned at an identical rate by all three models (CMS = 1.00), whereas SigNoz ranged from 0.00 to 0.67 across models and New Relic from 0.33 to 1.00. This supports treating Cross-Model Stability as a separate dimension rather than averaging models into a single score.

7.6 Limitations of This Pilot

- All three models originate from a single provider. Cross-Model Stability as measured here captures cross-version consistency within one model family, not cross-provider consistency as defined in Section 4.3. The reported value is an upper bound on what a genuinely cross-provider measurement would yield and should not be read as validating that dimension.
- One domain, one language, one day. The pilot covers six entities in a single English-language product category on a single date, with one response per prompt-model pair. It cannot speak to temporal drift, multilingual behaviour, or domain generality.
- The second annotator was a language model. Inter-rater agreement was estimated between the author and an LLM annotator applying the same written protocol, not between two independent human coders. The reported κ is evidence that the coding scheme is specifiable, not that it is reliable across human raters.
- The rescaling constant is approximate at small sample sizes. PR and CMS use the factor $1 - 2\sigma$, which assumes a maximum standard deviation of 0.5. For a binary variable over three observations the attainable

maximum is 0.471, so PR is compressed onto approximately [0.06, 1] rather than [0, 1] at this sample size. The distortion diminishes as the number of prompt variants grows, but studies with few variants should either report the exact bound or normalise by the attainable maximum.

- Test-retest was single-model and short-interval. Repeating the full prompt set on one model gave 89 percent agreement across 36 cells, both on exact endorsement codes and on binary mention. This indicates stability under immediate resampling; it says nothing about stability across model updates.

8. Discussion

8.1 Implications for GEO

The proposed framework shifts the focus of optimization from webpages to entities. This perspective aligns with the increasing role of conversational AI in information discovery and suggests that future optimization efforts may benefit from measuring entity-level visibility in addition to traditional search performance.

8.2 Practical Applications

Potential applications include: longitudinal monitoring of AI visibility, comparison of brands within the same industry, evaluation of optimization campaigns, assessment of multilingual visibility, and academic studies on generative information retrieval.

8.3 Limitations

Several limitations should be acknowledged. First, the framework has been demonstrated only through a small feasibility pilot and has not undergone comprehensive empirical validation; in particular, the pilot did not test cross-provider stability, the dimension most likely to require revision. Second, AI systems evolve rapidly, which may influence visibility independently of optimization efforts. Third, some dimensions, such as Recommendation Confidence, rely on operational definitions that require validation against human annotators rather than model annotators.

9. Future Research

Future work may extend the framework in several directions: empirical validation using large benchmark datasets, multilingual evaluation protocols, domain-specific weighting schemes, automated extraction of AIVI dimensions, integration with Retrieval-Augmented Generation evaluation, and longitudinal analyses of AI visibility over time.

10. Conclusion

The transition from document-centric search to AI-generated answers creates new challenges for measuring digital visibility. This paper introduced the Artificial Intelligence Visibility Index (AIVI), a conceptual framework designed to quantify entity visibility across Large Language Models.

By decomposing AI visibility into six observable dimensions and defining a standardized benchmark protocol, the framework provides a structured foundation for evaluating discoverability within conversational AI systems.

Rather than presenting a finalized metric, AIVI should be viewed as a research program: an initial proposal intended to stimulate discussion, empirical validation, and future standardization efforts in Generative Engine Optimization.

Declarations

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Conflicts of interest: The author declares no conflicts of interest.

Data availability: The prompt set, the coded observation matrix, and the computation script for the pilot reported in Section 7 are available from the author on request. The framework and benchmark protocol are fully specified in Sections 4 and 5.

Use of AI tools: Generative AI tools were used for language editing and formatting assistance. All conceptual content, framework design, and analysis are the author's own, and the author takes full responsibility for the content of this paper.

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About the author

İbrahim Göktaş is an independent researcher and consultant whose work focuses on AI visibility, Generative Engine Optimization (GEO), Answer Engine Optimization (AEO), and digital-authority strategy in the age of generative AI. This paper presents the foundational AIVI framework for measuring entity-level discoverability across Large Language Models.